ABSTRACT OF THE DISCLOSURE

A system and method for establishing a secure communication channel by using a temporary, session-specific, randomized data encoding map, effectively creating a form of physical layer encryption. The method comprises: securely sharing a cryptographic seed between two communication endpoints via a secure out-of-band channel; each endpoint independently generating an identical, pseudo-random encoding map based on the shared seed; using the generated map for data transmission over a high-cardinality parallel interface for a single communication session; and destroying the map upon session termination. Because the map linking data to physical signals is known only to the endpoints and exists only for the duration of the session, an eavesdropper intercepting the physical signals cannot decode the data without the map, rendering traditional decryption unnecessary for confidentiality.